NEJM AI

NEJM AI 2026;3(5)
DOI: 10.1056/AIdbp2500516

DATASETS, BENCHMARKS, AND PROTOCOLS

# EchoNext-Mini: A Dataset and Baseline AI Model for Detecting Structural Heart Disease from Electrocardiograms

John Weston Hughes, Ph.D.,[1,2] Linyuan Jing, Ph.D.,[3] Joshua Finer, M.S.,[3] Dustin Hartzel, B.S.,[3] Christopher Kelsey, B.S.,[3] Aaron Long, M.S.,[1,2] Daniel Rocha, M.A.,[3] Jeffrey Ruhl, M.S.,[3] Timothy Poterucha, M.D.,[1] and Pierre Elias, M.D.[1,2]

Received: May 13, 2025; Revised: January 29, 2026; Accepted: March 3, 2026; Published: April 16, 2026

## Abstract

**BACKGROUND** Structural heart disease (SHD) screening has traditionally been limited to echocardiography, limiting access and impacting patient outcomes. We recently published EchoNext, an artificial intelligence (AI) model for detecting structural heart disease from electrocardiograms (ECGs), creating new opportunities for screening.

**METHODS** In this work, we describe the EchoNext-Mini dataset and model. The EchoNext-Mini dataset is a deidentified subset of the EchoNext dataset consisting of 100,000 ECGs from NewYork-Presbyterian/Columbia University Irving Medical Center, along with corresponding echocardiogram-derived SHD diagnoses and demographic information. Using this dataset, we trained the EchoNext-Mini model, an AI model to detect structural heart disease from ECG waveforms with the same architecture as the original EchoNext model.

**RESULTS** The EchoNext-Mini dataset contains 100,000 ECGs from 36,286 patients along with 11 SHD diagnosis labels. Overall, 52% of ECGs are positive for any SHD, with the most common diagnoses being elevated left ventricular wall thickness (24%), heart failure with reduced ejection fraction (24%), and pulmonary hypertension (19%). The EchoNext-Mini model achieves an area under the receiver operating characteristic curve (AUROC) of 82.0 (80.9–83.1), 95% confidence interval (CI), a sensitivity of 70.1 (68.3–71.9), 95% confidence interval (CI), and a specificity of 77.9 (76.5–79.4) in detecting SHD from ECG waveforms, versus an AUROC of 85.2 achieved by the original EchoNext model. AUROCs in detecting subdiagnoses of structural heart disease range from 73.4 (71.9–75.1; left ventricular wall thickness ≥1.3 cm) to 86.6 (84.9–88.2; right ventricular systolic dysfunction).

**CONCLUSIONS** We have made the EchoNext-Mini model and dataset publicly available to enable further research in AI understanding of ECGs. The size of the dataset and prevalence of varied labels enables a wide range of research. The EchoNext-Mini model achieves performance similar to the larger EchoNext model and offers a strong benchmark on the EchoNext-Mini dataset.

*Timothy Poterucha and Pierre Elias contributed equally to this article.*

*The author affiliations are listed at the end of the article.*

*John Weston Hughes can be contacted at weh7014@nyp.org.*

NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

## Introduction

Structural heart disease (SHD), a broad category of diseases affecting the valves, walls, and chambers of the heart, affects tens of millions of patients[1] and accounts for hundreds of billions of dollars in health care expenditures in the United States each year.[2] Although prompt treatment of SHD leads to improved outcomes and quality of life, it remains underdiagnosed,[3] in large part because echocardiography — the gold-standard diagnostic test — can be costly and is not uniformly accessible. We recently published results on the EchoNext model along with the dataset on which it was developed and evaluated.[4]

EchoNext is a convolutional neural network model for detecting SHD from electrocardiogram (ECG) waveforms developed using data from the eight hospitals in the NewYork-Presbyterian medical system, achieving an area under the receiver operating characteristic curve (AUROC) of 85.2%, an area under the precision-recall curve (AUPRC) of 78.5% on an internal test set, and similar results on three external test sets. The model outperformed cardiologists in detecting SHD and showed favorable performance in a prospective silent validation study. The model also provided individual detection scores for 11 different subdiagnoses of SHD and shows impressive performance on those labels.

Open-source models and datasets have enabled innovation across the AI community for decades, and we aim to bring that same momentum to cardiology. To support this goal, we are sharing the EchoNext-Mini dataset — a subset of the primary EchoNext dataset — comprising 100,000 ECGs from NewYork-Presbyterian/Columbia University Irving Medical Center, along with curated labels for SHD diagnoses, ECG measurements, and patient demographic data. In addition, we provide the EchoNext-Mini model — a version of the EchoNext model trained on this dataset and achieving similar performance — as an open-source resource. In this work, we detail the development of the dataset and model, describe the dataset and model performance, and highlight potential future projects that the dataset enables. A summary of contributions is shown in Figure 1.

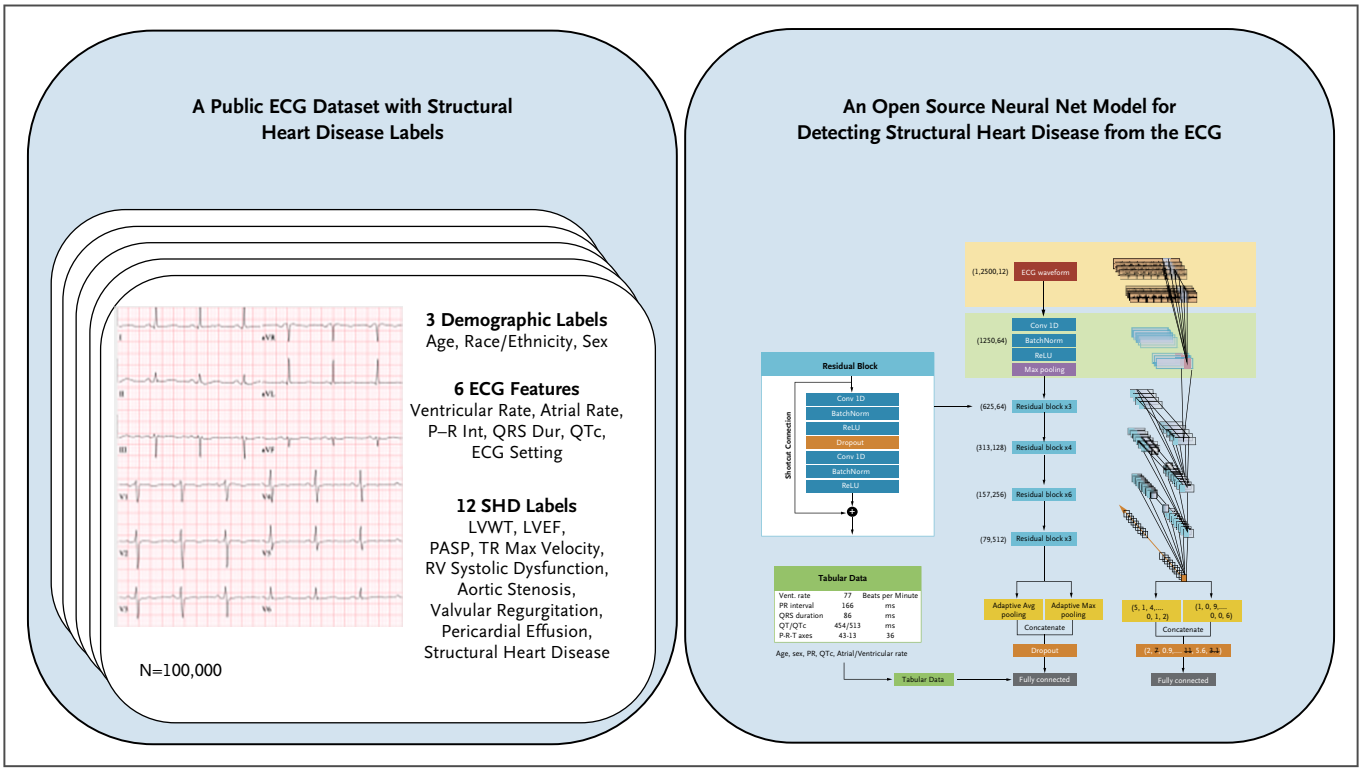



Figure 1. Summary of the Contributions of this Article.

ECG denotes electrocardiogram; Dur, duration; Int, interval; LVEF, left ventricular ejection fraction; LVWT, left ventricular wall thickness; PASP, pulmonary artery systolic pressure; QTc, corrected QT interval; SHD, structural heart disease; and TR Max Velocity, tricuspid regurgitation maximum velocity.



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

# Methods

## DATASET CURATION

The published EchoNext-Mini dataset comprises a sample of 100,000 ECGs and accompanying metadata and labels from the larger EchoNext dataset.[4] The 6,336,478 12-lead ECGs taken between December 2008 and December 2022 within the New York-Presbyterian hospital system were first filtered to remove records with missing measurements or age or sex information; patients under 18 years of age; ECGs failing the automated Method for Updating the Security Level Evaluation Instrument (MUSE) quality control check; and ECGs with ventricular pacing (Fig. 2). This resulted in 5,643,448 ECGs from 1,479,068 patients.

These ECGs were then matched with transthoracic echocardiograms (TTEs) without prosthetic valves to identify ECGs recorded prior to a TTE. ECGs obtained within 1 year before a TTE that confirmed a given diagnosis were labeled as positive for that diagnosis. ECGs preceding a TTE that did not show the diagnosis were labeled as negative, including ECGs recorded more than 1 year before the TTE, provided the patient never had that diagnosis on any TTE. Given the progressive nature of SHD diagnoses, patients generally only transition to more severe disease, so a negative label at any point in the future likely indicates a current negative label.

Granular measurements and labels were labeled "NaN" when there was more than a year between ECG and TTE, as those measurements may have changed between ECG and TTE dates. Previous analysis demonstrated that model performance is robust in windows between 3 months and 1 year, indicating that a 1-year window is acceptable.[4] This resulted in 1,245,273 ECGs with TTE-derived labels from 230,218 patients across 8 NewYork-Presbyterian hospitals. This dataset was used to train and evaluate the recently published EchoNext model.[4] From that dataset,

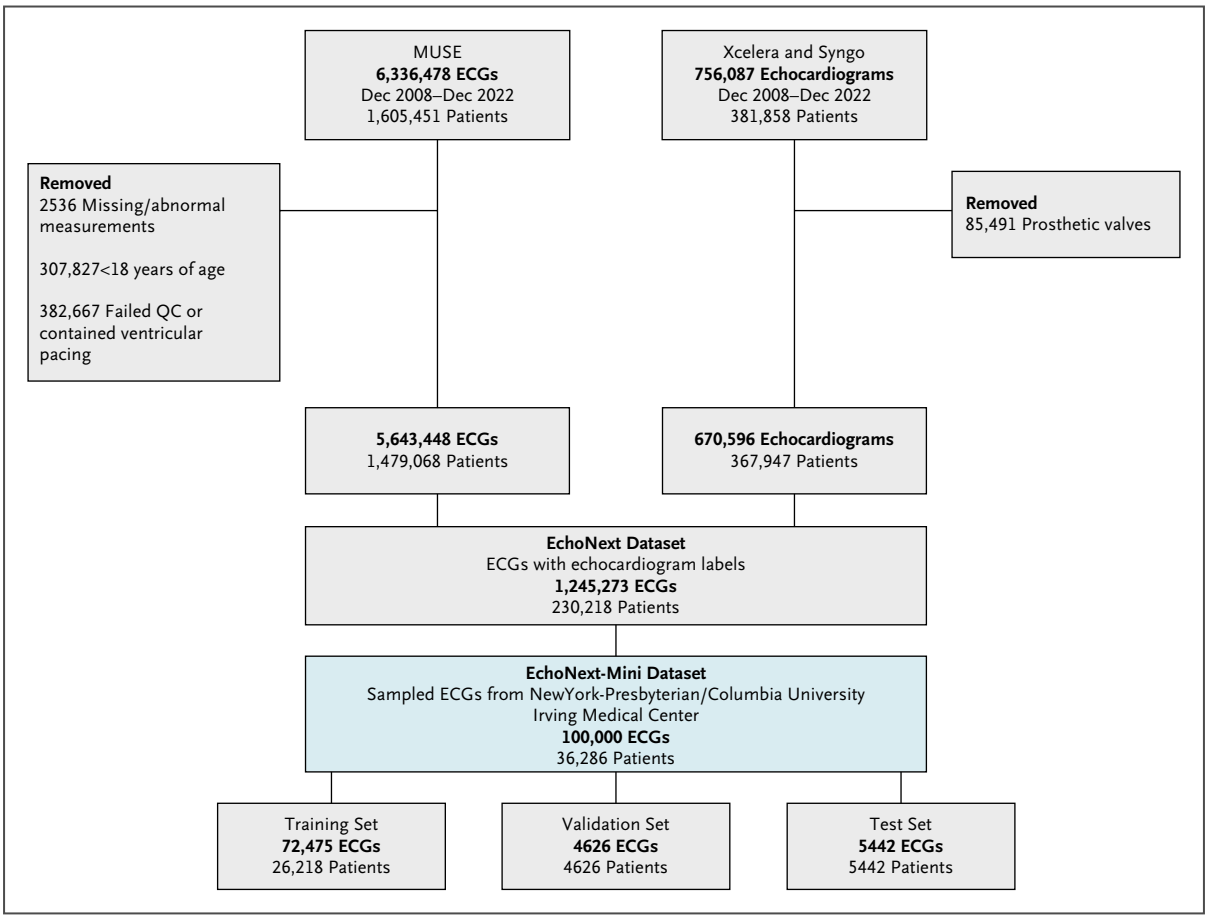



Figure 2. CONSORT flow diagram for the EchoNext-Mini Dataset.

CONSORT denotes Consolidated Standards of Reporting Trials; ECG, electrocardiogram; and MUSE, Method for Updating the Security Level Evaluation Instrument.



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

100,000 ECGs from the NewYork-Presbyterian/Columbia University Irving Medical Center site were randomly sampled to create the EchoNext-Mini dataset.

The dataset was also split at the patient level into training, validation, and test sets for model development, with only the most recent ECG for each patient being included in the validation and test sets, and no patients existed in multiple data splits. Thus, there could be multiple ECGs for one patient in the training set, while only a single ECG for each patient is included in the validation and test splits. Additional ECGs from patients in the validation and test sets were not used for model development or evaluation, but are included in the published dataset. The waveforms and metadata were stored in formats containing no identifiable information, and were assigned random patient identifiers. This retrospective study and data publication were approved by the Columbia University Institutional Research Board with waiver of patient consent.

## ELECTROCARDIOGRAM ACQUISITION

The ECGs were extracted from the GE HealthCare MUSE platform. Both waveforms and automated measurements were extracted for each record. Waveforms were recorded at 250 Hz for 10 seconds and consist of 12 leads of data stored as 12 channels of 2500 measurements each. Measurements included ventricular and atrial rates, P–R interval, QRS duration, and Bazett's corrected QT interval (QTc), all of which were evaluated automatically using the MUSE software and not human corrected. ECG setting was also extracted from ECG data (i.e., inpatient, outpatient, emergency, or procedural, mapped from individual ECG cart location). Records were also joined with patient metadata from the electronic health record, including age at the time of ECG, sex, and race or ethnicity. Ages greater than 90 years were set to 90 to preserve patient privacy. ECG waveforms were filtered using a per-lead median signal filter, clipped at the dataset-wide 0.1st and 99.9th percentile voltages, and normalized dataset-wide on a per-lead basis using mean and standard deviation.

## ECHOCARDIOGRAM LABELS

Echocardiogram measurements at NewYork-Presbyterian/Columbia University Irving Medical Center were extracted from the Siemens Healthineers Syngo Dynamics platform from cardiologists' overreads of ultrasounds performed during clinical care. Five numerical values were extracted: left ventricular ejection fraction (LVEF), interventricular septum thickness, posterior wall thickness, pulmonary artery systolic pressure (PASP), and tricuspid regurgitation maximum velocity (TR max velocity). The maximum of the two wall thicknesses was considered the left ventricular wall thickness (LVWT). In addition, six categorical diagnoses were extracted: aortic, mitral, tricuspid, and pulmonic regurgitation (given a value of none/trace, mild, moderate, or severe), right ventricular systolic function (given a value of normal, mildly reduced, moderately reduced, or severely reduced), pericardial effusion (given a value of none, trace, small, moderate, or large), and aortic stenosis (given a value of none/trace, mild, moderate, or severe). Labels were then binarized to create designations for moderate or greater disease. Continuous labels were binarized to moderate disease at LVWT≥1.3, LVEF≤45, PASP≥45, and TR maximum velocity≥3.2. To be included in the study, an echocardiography report was required to include LVEF and at least one relevant valve finding. Missing categorical and numerical labels were binarized to negative diagnoses. For example, if an echocardiogram report made no mention of aortic stenosis and did not evaluate peak aortic velocity, the echocardiogram was assumed not to contain aortic stenosis. Physician-led chart review of 50 cases for each missing value verified that the lack of fine-grained labels provided in the TTE report implied that the disease was not present, and such cases were collapsed to the "normal" or "none" categories. None of the labels reviewed had more than 1 out of 50 missing labels that were not "normal" or "none." All echocardiography labels were established following relevant clinical guidelines from the American Society of Echocardiography.[5-7] Both binarized labels (used for training and evaluating the model and useful in general for classification tasks) and continuous/categorical labels (at the highest level of detail available) are published in the data release.

## MODEL DEVELOPMENT

The EchoNext-Mini model was trained and evaluated using the EchoNext-Mini dataset, closely following the protocol and architecture used to train the primary EchoNext Model[4] and the ValveNet Model.[8] The model takes the full ECG as input, along with key automatic measurements from ECGs and demographic features. It was trained in a multitask setup to detect both the presence of each of the 11 individual SHD subdiagnosis labels included in the EchoNext-Mini dataset and the composite label indicating ECGs aligned with any of the subdiagnoses. The model architecture is a single-lead ResNet similar in structure to the ResNet-34 vision model, but taking a 12-lead 1D time



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

series instead of a 3-channel image as input. Convolutional kernels at lower levels learn low-level patterns that repeat across an ECG, while higher layers learn to synthesize these features into high-level patterns and diagnoses. The final layer of the model outputs 12 logits, 1 detecting composite SHD, and the others are the 11 SHD subdiagnoses. Each is trained with an independent binary cross-entropy loss and outputs a score between 0 and 1. A summary of the model is shown in Figure 1, and full details are shown in Table S1 in the Supplementary Appendix. Tabular information is added to the final linear layers using late fusion. Missing continuous variables were imputed to the median value for that variable, except for atrial rate and P–R interval, which were set to 0; there was no missingness in age or sex. The median imputations were computed by taking the median of all nonmissing values for a given field in the training set (e.g., a missing QTc value is set to the median of all nonmissing QTc values in the training set). During validation and testing, data were imputed using medians fit on the training set to mimic real-world deployment. Optimal hyperparameters from the primary EchoNext model were used, including a batch size of 16, a filter size of 16, a weight decay of 0.01, a dropout of 0.5, and an Adam learning rate of 5e-5. The model was developed using PyTorch and Python.

### STATISTICAL ANALYSIS

Model performance was assessed using the receiver operating characteristic curve, the precision-recall curve, and the areas under each curve (AUROC, AUPRC). In addition, binary prediction metrics (sensitivity, specificity, positive predictive value, and negative predictive value) were computed at the cutoff, leading to 70% sensitivity. To understand model performance across demographic groups, the AUROC was assessed for different racial, ethnic, and sex subgroups. In addition, performance was assessed in four different clinical contexts (inpatient, outpatient, emergency, and procedural) to quantify expected performance when applied to different use cases and care patterns. For all performance values, 95% confidence intervals were computed via bootstrapping with 1000 iterations. Performance is reported both on the entire dataset and in subsets of the population based on demographics and clinical settings to assess model robustness across populations. Statistical significance of differences in subgroup performance was assessed via confidence intervals. The dataset was summarized using means and interquartile ranges. All analyses and training were performed using Python and the scikit-learn PyTorch packages.

## Results

### THE ECHONEXT-MINI DATASET

The EchoNext-Mini dataset consists of 100,000 ECGs from 36,286 patients. The training, validation, and test sets consist of 72,475 (26,218), 4626 (4626), and 5442 (5442) ECGs (patients), respectively. The average patient was 63 years of age (interquartile range: 51–73) at the time of ECG and 53.6% were male. The population is diverse, with 31% of ECGs being from Hispanic patients, 29.2% from non-Hispanic white patients, 16.5% from Black patients, 3.4% from Asian patients, and 19.9% from patients of other or unknown races or ethnicities. ECG sources are varied, with 48.2% originating from inpatient settings, 31.5% from emergency settings, 17.2% from outpatients, and 3.2% from procedural settings. A full summary of the demographics, measurements, and diagnoses in the dataset is available in Table 1. Missingness of data is described in Tables S2 to S4. Most variables were available for all ECGs, while the atrial rate was missing in 0.6% of cases and the P–R interval was missing in 10.4% of cases.

Most patients in the dataset (52.2%) exhibit some form of moderate or greater SHD. The most common diagnosis is LVWT≥1.3, with 24.2% of patients, followed by LVEF≤45 (24.0%), PASP≥45 (19.0%), right ventricular systolic dysfunction (13.2%), and tricuspid regurgitation (10.7%). In addition, 27.7% exhibit severe SHD, with 16.5% having LVEF≤35 and 6.6% having a PASP≥60. The rarest diagnosis was severe pulmonary regurgitation, with only 52 examples out of 100,000. The median LVEF was 57.5%, with half of the patients falling between 45% and 62.5%. Missingness of raw numerical and categorical echocardiogram labels varied between 8.9% and 55.0% (Table S2). Summaries of all binarized labels are shown in Figure 3, and pairwise scatter plots of continuous values are shown in Figure S1.

Several of the SHD diagnoses show high correlation (Fig. S2). The labels PASP≥45 and TR Max Velocity≥3.2 have a Matthews correlation coefficient (MCC) of 0.65; PASP≥45 and tricuspid regurgitation have an MCC of 0.44; and TR Max Velocity≥3.2 and tricuspid regurgitation have an MCC of 0.4. Right ventricular systolic dysfunction is also strongly correlated with LVEF≤45 (0.40) and tricuspid regurgitation (0.36). There were no strong negative correlations between any labels. All diagnoses had high odds ratios with one another, ranging between 13.83 and 23.26 (Fig. S3).



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

**Table 1. Demographics and Diagnoses of the EchoNext-Mini Dataset.***

| Variable | All ECGs | Latest ECG | Training Set | Validation Set | Test Set |
|---|---|---|---|---|---|
| Number of samples | 100,000 | 36,286 | 72,475 | 4626 | 5442 |
| Number of patients | 36,286 | 36,286 | 26,218 | 4626 | 5442 |
| Male — no. (%) | 53,581 (53.6%) | 17,953 (49.5%) | 38,951 (53.7%) | 2270 (49.1%) | 2711 (49.8%) |
| LVEF≤45 cm — no. (%) | 23,892 (24.0%) | 6535 (18.0%) | 16,962 (23.4%) | 866 (18.7%) | 962 (17.7%) |
| LVWT≥1.3 cm — no. (%) | 24,220 (24.2%) | 6828 (18.8%) | 17,667 (24.4%) | 877 (19.0%) | 1061 (19.5%) |
| Moderate or greater aortic stenosis — no. (%) | 4054 (4.1%) | 1973 (5.4%) | 2919 (4.0%) | 252 (5.4%) | 286 (5.3%) |
| Moderate or greater aortic regurgitation — no. (%) | 1264 (1.3%) | 433 (1.2%) | 878 (1.2%) | 62 (1.3%) | 66 (1.2%) |
| Moderate or greater mitral regurgitation — no. (%) | 8451 (8.5%) | 2352 (6.5%) | 6137 (8.5%) | 282 (6.1%) | 337 (6.2%) |
| Moderate or greater tricuspid regurgitation — no. (%) | 10,651 (10.7%) | 2534 (7.0%) | 7707 (10.6%) | 305 (6.6%) | 353 (6.5%) |
| Moderate or greater pulmonary regurgitation — no. (%) | 821 (0.8%) | 156 (0.4%) | 603 (0.8%) | 21 (0.5%) | 20 (0.4%) |
| Moderate or greater RV systolic dysfunction — no. (%) | 13,243 (13.2%) | 2899 (8.0%) | 9597 (13.2%) | 368 (8.0%) | 419 (7.7%) |
| Moderate–large pericardial effusion — no. (%) | 3023 (3.0%) | 475 (1.3%) | 2079 (2.9%) | 52 (1.1%) | 69 (1.3%) |
| PASP≥45 mmgh — no. (%) | 18,993 (19.0%) | 4685 (12.9%) | 13,727 (18.9%) | 581 (12.6%) | 699 (12.8%) |
| TR max velocity≥3.2 m/s — no. (%) | 10,212 (10.2%) | 2365 (6.5%) | 7492 (10.3%) | 267 (5.8%) | 375 (6.9%) |
| Moderate or greater structural heart disease — no. (%) | 52,188 (52.2%) | 15,703 (43.3%) | 37,958 (52.4%) | 1990 (43.0%) | 2318 (42.6%) |
| Age (years) | 63 (51, 73) | 64 (52, 75) | 63 (52, 73) | 64 (52, 75) | 64 (52, 74) |
| Ventricular rate (%) | 81 (69, 95) | 78 (66, 92) | 81 (69, 95) | 78 (66, 92) | 78 (66, 92) |
| Atrial rate (bpm) | 82 (69, 97) | 78 (66, 93) | 81 (69, 97) | 78 (66, 93) | 78 (66, 94) |
| P–R interval (ms) | 154 (138, 174) | 156 (140, 176) | 154 (138, 174) | 156 (140, 176) | 156 (140, 176) |
| QRS duration (ms) | 90 (80, 102) | 88 (80, 100) | 90 (80, 102) | 88 (80, 100) | 88 (80, 100) |
| QTc | 449 (427, 474) | 444 (423, 468) | 449 (427, 474) | 445 (423, 469) | 444 (423, 467) |
| QT interval (ms) | 389.2 (359.1, 421.0) | 392.7 (363.1, 423.1) | 389.4 (359.3, 421.2) | 391.6 (363.1, 423.4) | 391.8 (362.8, 422.5) |
| LVEF (%) | 57.5 (45.0, 62.5) | 57.5 (52.0, 62.5) | 57.5 (45.0, 62.5) | 57.5 (50.0, 62.5) | 57.5 (52.5, 62.5) |
| Intraventricular septum thickness (cm) | 1.1 (1.0, 1.2) | 1.0 (0.9, 1.2) | 1.1 (1.0, 1.2) | 1.0 (0.9, 1.2) | 1.0 (0.9, 1.2) |
| Left ventricular posterior wall thickness (cm) | 1.0 (0.9, 1.2) | 1.0 (0.9, 1.1) | 1.0 (0.9, 1.2) | 1.0 (0.9, 1.1) | 1.0 (0.9, 1.1) |
| Left ventricular wall thickness (cm) | 1.1 (1.0, 1.3) | 1.1 (0.9, 1.2) | 1.1 (1.0, 1.3) | 1.1 (0.9, 1.2) | 1.1 (0.9, 1.2) |
| PASP | 38 (31, 49) | 36 (29, 45) | 38 (31, 49) | 36 (29, 45) | 36 (29, 45) |
| TR max velocity (m/s) | 2.7 (2.3, 3.1) | 2.6 (2.3, 3.0) | 2.7 (2.3, 3.1) | 2.6 (2.3, 3.0) | 2.6 (2.3, 3.0) |
| Race or ethnicity — no. (%)† | | | | | |
| Hispanic | 31,013 (31.0%) | 11,006 (30.3%) | 22,806 (31.5%) | 1351 (29.2%) | 1649 (30.3%) |
| White | 29,211 (29.2%) | 10,635 (29.3%) | 21,289 (29.4%) | 1385 (29.9%) | 1569 (28.8%) |
| Black | 16,461 (16.5%) | 5424 (14.9%) | 11,559 (15.9%) | 728 (15.7%) | 846 (15.5%) |
| Unknown | 12,458 (12.5%) | 5159 (14.2%) | 8947 (12.3%) | 648 (14.0%) | 768 (14.1%) |
| Other | 7429 (7.4%) | 2980 (8.2%) | 5272 (7.3%) | 380 (8.2%) | 457 (8.4%) |
| Asian | 3428 (3.4%) | 1082 (3.0%) | 2602 (3.6%) | 134 (2.9%) | 153 (2.8%) |
| ECG setting — no. (%) | | | | | |
| Inpatient | 48,162 (48.2%) | 14,932 (41.2%) | 34,906 (48.2%) | 1903 (41.1%) | 2203 (40.5%) |
| Emergency | 31,453 (31.5%) | 13,026 (35.9%) | 22,811 (31.5%) | 1688 (36.5%) | 1971 (36.2%) |
| Outpatient | 17,194 (17.2%) | 6846 (18.9%) | 12,423 (17.1%) | 858 (18.5%) | 1059 (19.5%) |
| Procedural | 3191 (3.2%) | 1482 (4.1%) | 2335 (3.2%) | 177 (3.8%) | 209 (3.8%) |

* Numerical values are summarized as median (25th percentile, 75th percentile). ECG denotes electrocardiogram; LVEF, left ventricular ejection fraction; LVWT, left ventricular wall thickness; PASP, pulmonary artery systolic pressure; QTc, corrected QT interval; RV, right ventricle; and TR max velocity, tricuspid regurgitation maximum velocity.

† Race and ethnicity were abstracted from the electronic medical record. Other includes patients with a race/ethnicity of other in the electronic health record, as well as Native Hawaiian/Pacific Islander (N=297) and Native American/Alaskan (N=144).



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

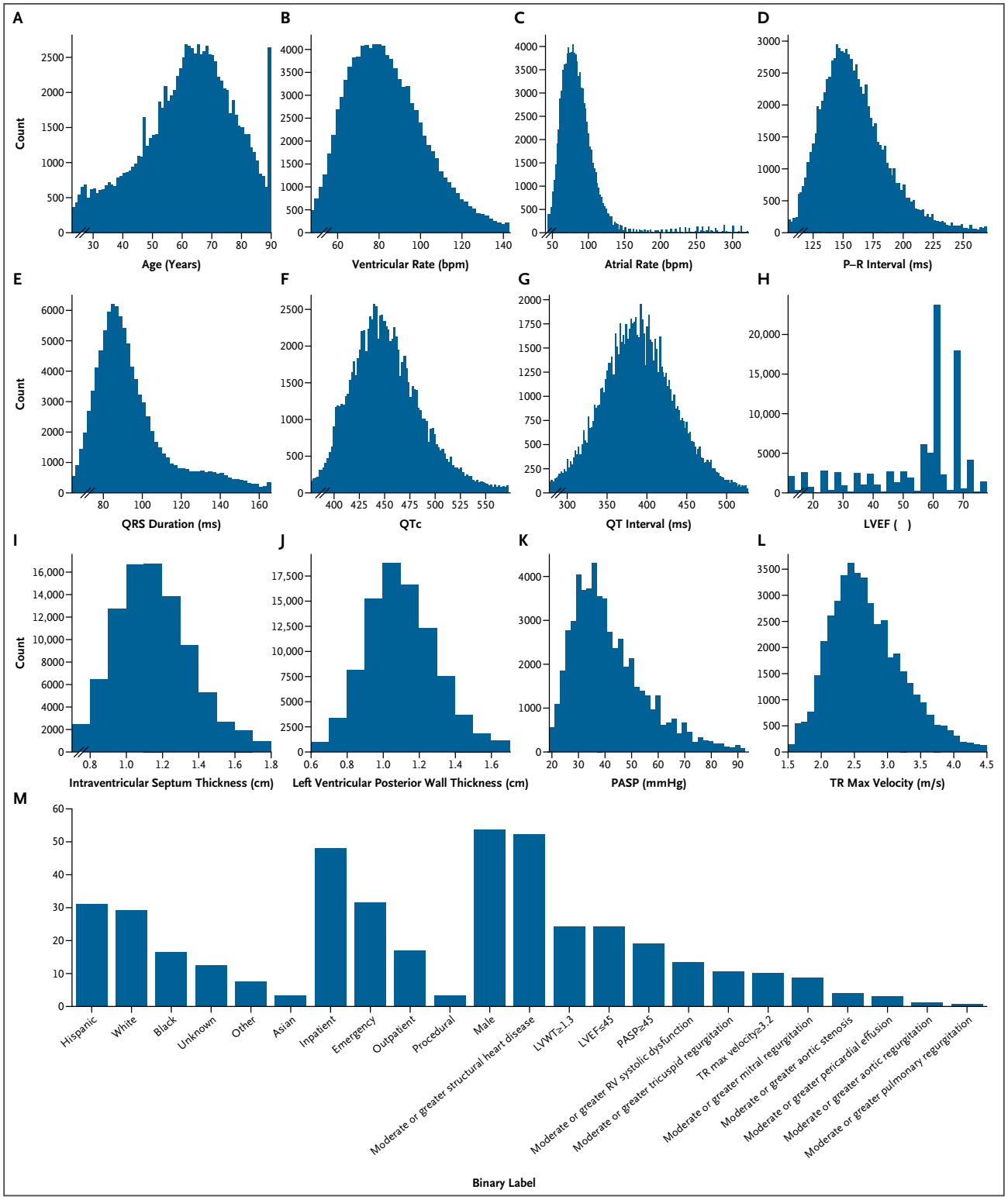



Figure 3. Histograms of Continuous Measurements and Proportion Positive for Each Binary Label in the EchoNext-Mini Dataset.

Panels A to L are histograms of continuous measurements in the EchoNext-Mini dataset. Panel M shows the proportion positive for each binary label. LVEF denotes left ventricular ejection fraction; LVWT, left ventricular wall thickness; PASP, pulmonary artery systolic pressure; QTc, corrected QT interval; and TR max velocity, tricuspid regurgitation maximum velocity.



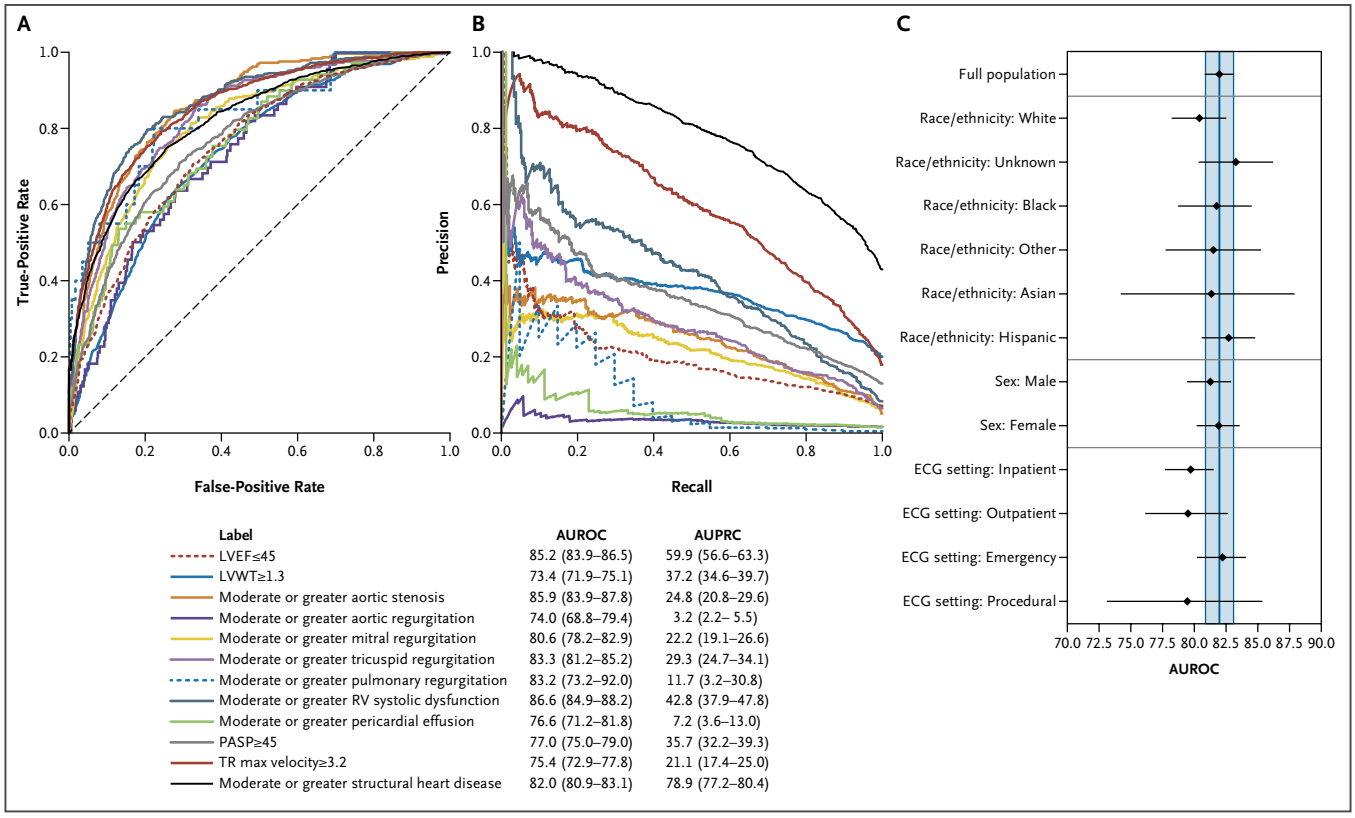
NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

## THE ECHONEXT-MINI MODEL

Using the EchoNext-Mini dataset, the EchoNext-Mini model was trained to detect SHD from ECG and demographic data. The model consists of 1,046,080 parameters across 34 layers.

Model performance is summarized in Figure 4, and is similar to the primary EchoNext model, which was trained on the primary EchoNext dataset.[4] In detecting the composite SHD class (including ECGs with any subdiagnosis of SHD), EchoNext-Mini achieves an AUROC of 82.0 (95% confidence interval, 80.9 to 83.1) and an AUPRC of 78.9 (77.2 to 80.4). At a model cutoff value selected to allow for a sensitivity of 70%, the model achieved a sensitivity of 70.1 (68.3–71.9), a specificity of 77.9 (76.5–79.4), a positive predictive value of 70.2 (68.3–72.0), and a negative predictive value of 77.8 (76.4–79.3) in predicting SHD. Individual SHD diagnoses are well detected, with AUROCs ranging between 73.4 (71.9–75.1; LVWT≥1.3) and 86.6 (84.9–88.2; right ventricular systolic dysfunction). Individual AUPRCs are lower, ranging between 3.2 (2.2–5.5; aortic stenosis) and 59.9 (56.6–63.3; LVEF≤45), reflecting both the lower prevalence of some diagnoses and the challenge of distinguishing between different SHD subdiagnoses.

We also assessed the performance of the model in detecting SHD in various subgroups of the EchoNext-Mini dataset based on demographics and clinical setting (Fig. 4). There was no group in which the model achieved a significantly different AUROC versus the entire population. Comparing different racial and ethnic groups, the model performed worst on white patients, achieving an AUROC of 80.4 (78.3–82.5), and best on the group of patients




| Label | AUROC | AUPRC |
|---|---|---|
| LVEF≤45 | 85.2 (83.9–86.5) | 59.9 (56.6–63.3) |
| LVWT≥1.3 | 73.4 (71.9–75.1) | 37.2 (34.6–39.7) |
| Moderate or greater aortic stenosis | 85.9 (83.9–87.8) | 24.8 (20.8–29.6) |
| Moderate or greater aortic regurgitation | 74.0 (68.8–79.4) | 3.2 (2.2– 5.5) |
| Moderate or greater mitral regurgitation | 80.6 (78.2–82.9) | 22.2 (19.1–26.6) |
| Moderate or greater tricuspid regurgitation | 83.3 (81.2–85.2) | 29.3 (24.7–34.1) |
| Moderate or greater pulmonary regurgitation | 83.2 (73.2–92.0) | 11.7 (3.2–30.8) |
| Moderate or greater RV systolic dysfunction | 86.6 (84.9–88.2) | 42.8 (37.9–47.8) |
| Moderate or greater pericardial effusion | 76.6 (71.2–81.8) | 7.2 (3.6–13.0) |
| PASP≥45 | 77.0 (75.0–79.0) | 35.7 (32.2–39.3) |
| TR max velocity≥3.2 | 75.4 (72.9–77.8) | 21.1 (17.4–25.0) |
| Moderate or greater structural heart disease | 82.0 (80.9–83.1) | 78.9 (77.2–80.4) |

Figure 4. Receiver Operating Characteristic and Precision Recall Curves for the EchoNext-Mini Dataset.

Panel A shows receiver operating characteristic curves of performance of the EchoNext-Mini model for each label in the EchoNext-Mini dataset, along with areas under the receiver operating characteristic curve. Panel B shows precision-recall curves of performance of the EchoNext-Mini model for each label in the EchoNext-Mini dataset, along with areas under the precision-recall curves. Panel C shows areas under the receiver operating characteristic curve and confidence intervals of the EchoNext-Mini model in predicting moderate or greater structural heart disease in different demographic and clinical subgroups. The light blue band shows the confidence interval for model performance in the entire population. AUPRC denotes area under the precision-recall curve; AUROC, area under the receiver operating curve; ECG, electrocardiogram; LVEF, left ventricular ejection fraction; LVWT, left ventricular wall thickness; PASP, pulmonary artery systolic pressure; RV, right ventricle; and TR max velocity, tricuspid regurgitation maximum velocity.



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

whose race and ethnicity were unknown, with an AUROC of 83.3 (80.4–86.1). Male and female patients performed similarly. When restricted to a single clinical setting, performance declined slightly — likely due to shifts in prevalence — though these decreases did not reach statistical significance. The emergency department was the sole exception, exhibiting a modest, nonsignificant increase in performance. Overall, the model demonstrated strong and robust performance across all subgroups.

## Discussion

We present the EchoNext-Mini dataset and model, a new public dataset and open model for detecting SHD from ECGs generated during the development of the larger EchoNext dataset and model.[4] Although the EchoNext-Mini dataset is only a fraction of the size of the primary EchoNext dataset, the EchoNext-Mini model performs competitively with the primary EchoNext model. We hope that publishing the dataset will spur continued research and development of new AI tools for ECG modeling.

### FURTHER RESEARCH USING THE ECHONEXT-MINI DATASET

There are several exciting avenues for further exploration of the EchoNext-Mini dataset. First, while the published EchoNext-Mini model is a strong baseline, we expect that further research into AI ECG models could lead to increased performance in detecting SHD. For instance, while our model development exploration focused on convolutional neural net architectures, transformer or other architectures might achieve higher performance in sequence classification. The analysis of model performance in demographic subgroups found limited differences in model performance between groups; however, other recent work has shown changes in model performance when looking at intersections between groups, and the diversity of this dataset makes it a good fit for similar analyses.[9] Twelve-lead ECG datasets are also excellent for performing analyses of the relative importance of individual leads, with applications to wearable devices.[10] Our results demonstrate that strong performance is achievable in detecting SHD subdiagnoses individually and as a composite label, but it is less clear whether it is possible to distinguish between SHD subdiagnoses using ECGs; labels are available in this dataset to further examine this problem. The dataset can also be used in conjunction with other ECG datasets, such as MIMIC-IV-ECG[11] and PTB-XL,[12] to evaluate pretraining, for fine-tuning, and to transfer learning performance of models across datasets. It can also be used in training foundation models. Finally, the dataset contains multiple ECGs for many patients, presenting an opportunity to research how changes in ECG results inform SHD diagnosis.

### FURTHER RESEARCH USING THE ECHONEXT-MINI MODEL

The EchoNext-Mini model likewise presents several potential research opportunities. The model can serve as a baseline for any project involving the detection of SHD. The risk of SHD output by the model can be used as a feature for downstream models and workflows, whether they be other AI systems, more traditional statistical analyses, decision-making processes, or other types of research. Together, the model and dataset can be used for model-based interpretability research for ECGs, which is currently lacking.[13]

The EchoNext-Mini dataset and model have several unique features that make them an important contribution to the open ECG dataset space. The dataset is highly diverse, coming from the NewYork-Presbyterian/Columbia University Irving Medical Center, which serves a diverse neighborhood of northern Manhattan, with 31.0% Hispanic patients and 16.5% Black patients. Given the known differences in ECG physiology[14,15] and evidence of underdiagnosis of heart disease[16] in Black and female patients, more diverse datasets are highly valuable. Other large ECG datasets have been open-sourced recently, including the CODE-15% dataset[17] of 345,779 examples and the MIMIC-IV-ECG dataset[11] of approximately 800,000 examples. Our work supplements and builds on these publications. The latter dataset could be merged with the 4579 patients in the MIMIC-IV Echocardiogram database, with additional parsing and quality control. To our knowledge, EchoNext-Mini is the first digitized ECG dataset with several thousand SHD labels and curated SHD diagnosis labels, and the first open-source model released to detect SHD. The availability of more granular sublabels of SHD offers yet greater potential.

As with all datasets and models, EchoNext-Mini has some drawbacks. The numerical ECG features were extracted automatically from ECGs using standard software, which is known to be error-prone, especially in the case of the QTc.[18] In addition, there is missingness in some of the extracted ECG features, including P–R intervals and atrial rates, which could diminish model performance. Similarly, there is missingness in echocardiogram labels due to incomplete reports, which might lead to some inaccuracy during training and evaluation. Although the chart review found that over 98% of missing labels are true-negative diagnoses, a minority of these cases may be abnormal cases with labels



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

missing. While the dataset had high numbers of Hispanic and Black patients, it consisted of only 3.4% Asian patients by ECG. The dataset consists of data from only a single institution, leading to potential biases in diagnosis and degradation of performance when applied at other centers. All the ECGs in the dataset were recorded using machines from the same vendor, further increasing the possibility of generalization issues.

# Conclusion

This work introduces two resources: the EchoNext-Mini dataset and model. A thorough description of the dataset's development and characteristics is provided, including a correlation analysis of the labels. Evaluation of model performance demonstrates strong results, including robustness across various demographic and clinical subgroups. Practical examples illustrate how the data and model can be used in practice to facilitate future research. Open datasets and models have driven innovation in AI, and with this release, we hope to bring that momentum to the ECG domain. We look forward to seeing further discovery and development using these resources.

## Disclosures

Author disclosures and other supplementary materials are available at ai.nejm.org.

The EchoNext-Mini dataset and model are both publicly available to spur additional research and innovation. The dataset is available at https://physionet.org/content/echonext/, with the DOI https://doi.org/10.13026/r9pp-3y42, along with a description of the file structure. Model weights are available at https://github.com/PierreElias/IntroECG/tree/master/7-EchoNext%20Minimodel, including parameters for waveform normalization and tabular data normalization and imputation.

Example code is available in the same GitHub repository as the weights. There is a Docker image allowing for end-to-end running of the EchoNext-Mini model on diverse computational setups, as well as snippets for running each step of the preprocessing and inference pipeline individually. There are explicit examples for running on the EchoNext-Mini dataset, as well as instructions for how to format other data to be input into the model. In addition, there is an IPython notebook showing how to load the EchoNext-Mini dataset and create each figure and table in this article.

## Author Affiliations

[1] Seymour, Paul, and Gloria Milstein Division of Cardiology, Department of Medicine, Columbia University Irving Medical Center, New York, NY, USA

[2] Department of Biomedical Informatics, Columbia University, New York, NY, USA

[3] NewYork-Presbyterian Hospital, New York, NY, USA

## References

1. Tsao CW, Aday AW, Zlmarzooq ZI, et al. Heart disease and stroke statistics-2023 update: a report from the American Heart Association. Circulation 2023;147:e93-e621. DOI: 10.1161/CIR.0000000000001123.
2. Mallow PJ, Chen J, Moore M, Gunnarsson C, Rizzo JA. Incremental direct healthcare expenditures of valvular heart disease in the USA. J Comp Eff Res 2019;8:879-887. DOI: 10.2217/cer-2019-0007.
3. Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. Circulation 2021;143:e72-e227. DOI: 10.1161/CIR.0000000000000923.
4. Poterucha TJ, Jing L, Ricart RP, et al. Detecting structural heart disease from electrocardiograms using AI. Nature 2025;644:221-230. DOI: 10.1038/s41586-025-09227-0.
5. Baumgartner H, Hung J, Bermejo J, et al. Recommendations on the echocardiographic assessment of aortic valve stenosis: a focused update from the European Association of Cardiovascular Imaging and the American Society of Echocardiography. J Am Soc Echocardiogr 2017;30:372-392. DOI: 10.1016/j.echo.2017.02.009.
6. Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. Eur Heart J Cardiovasc Imaging 2015;16:233-271. DOI: 10.1093/ehjci/jev014.
7. Mitchell C, Rahko PS, Blauwet LA, et al. Guidelines for performing a comprehensive transthoracic echocardiographic examination in adults: recommendations from the American Society of Echocardiography. J Am Soc Echocardiogr 2019;32:1-64. DOI: 10.1016/j.echo.2018.06.00.
8. Elias P, Poterucha TJ, Rajaram V, et al. Deep learning electrocardiographic analysis for detection of left-sided valvular heart disease. J Am Coll Cardiol 2022;80:613-626. DOI: 10.1016/j.jacc.2022.05.029.
9. Kaur D, Eston Hughes J, Rogers AJ, et al. Race, sex, and age disparities in the performance of ECG deep learning models predicting heart failure. Circ Heart Fail 2024;17:e010879. DOI: 10.1161/CIRCHEARTFAILURE.123.010879.
10. Chiu I-M, Wu P-J, Zhang H, et al. Serum potassium monitoring using AI-enabled smartwatch electrocardiograms. JACC Clin Electrophysiol 2024;10:2644-2654. DOI: 10.1016/j.jacep.2024.07.023.
11. Gow B, Pollard T, Nathanson LA, et al. MIMIC-IV-ECG: diagnostic electrocardiogram matched subset. PhysioNet. September 15, 2023 (https://doi.org/10.13026/4NQG-SB35).
12. Wagner P, Strodthoff N, Bousseljot R-D, et al. PTB-XL, a large publicly available electrocardiography dataset. Sci Data 2020;7:154. DOI: 10.1038/s41597-020-0495-6.



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.

13. Hughes JW, Somani S, Elias P, et al. Simple models versus deep learning in detecting low ejection fraction from the electrocardiogram. Eur Heart J Digit Health 2024;5:427-434. DOI: 10.1093/ehjdh/ztae034.

14. Macfarlane PW, Katibi IA, Hamde ST, et al. Racial differences in the ECG — selected aspects. J Electrocardiol 2014;47:809-814. DOI: 10.1016/j.jelectrocard.2014.08.003.

15. Kornberg B. S-T segment variants in young, adult, black women. J Am Osteopath Assoc 1983;83:35-38.

16. Sandhu AT, Tisdale RL, Rodriguez F, et al. Disparity in the setting of incident heart failure diagnosis. Circ Heart Fail 2021;14:e008538. DOI: 10.1161/CIRCHEARTFAILURE.121.008538.

17. Lima EM, Ribeiro AH, Paixão GMM, et al. Deep neural network-estimated electrocardiographic age as a mortality predictor. Nat Commun 2021;12:5117. DOI: 10.1038/s41467-021-25351-7.

18. Neumann B, Vink AS, Hermans BJM, et al. Manual versus automatic assessment of the QT-interval and corrected QT. Europace 2023;25:euad213. DOI: 10.1093/europace/euad213.



NEJM AI is produced by NEJM Group, a division of the Massachusetts Medical Society.
Downloaded from ai.nejm.org on May 22, 2026.